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לחינוך סייבר
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# Reverse Engineering Course

## AntiDebug Techniques

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# AntiDebug Techniques

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- We shall discuss only FEW AntiDebug methods:
  - IsBeingDebugged
  - NtGlobalFlag
  - CheckRemoteDebuggerPresent
  - Check for breakpoints
  - Check code integrity
  - Monitor thread
  - TLS Callback
  - Packed file
  - https://anti-debug.checkpoint.com/

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# PEB

- Process Environment Block – windows data structure
  - We shall use 32 bit PEB values
  - https://learn.microsoft.com/en-us/windows/win32/api/winternl/ns-winternl-peb

- Set eax to 0x30
- Patch code somewhere: mov eax, fs:[eax]

# IsBeingDebugged

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- IsBeingDebugged –at $PEB + 2
- Can be called by IsDebuggerPresent
- How to bypass?
  - Edit the PEB
  - Simply jump over IsDebuggerPresent
  - Patch code (change conditions etc)

# NtGlobalFlag

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- $PEB + 0x68
- A debugger will usually set the flags:
  - `FLG_HEAP_ENABLE_TAIL_CHECK (0x10)`
  - `FLG_HEAP_ENABLE_FREE_CHECK (0x20)`
  - `FLG_HEAP_VALIDATE_PARAMETERS (0x40)`

# CheckRemoteDebuggerPresent

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- https://learn.microsoft.com/en-us/visualstudio/debugger/remote-debugger-port-assignments?view=vs-2022
- Checks if a process has a non zero debug port
- How to bypass?
  - Same idea as IsDebuggerPresent

# Check Breakpoints

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- Debuggers replace a byte with “CC” (int 3)
- Demo:

```
xor     eax, eax ; breakpoint here
mov     al, [0x402000]
```

- Search for “CC” opcode in the text section

# Check Code Integrity

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- Pre-Calculate the sum (or other hash function) of all bytes
- Test during runtime that value is same

# Monitor Thread

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- https://www.youtube.com/watch?v=WFSJok0AceQ&ab_channel=FILMCRUX
- Demo – AntiDebug.exe

# Hands On

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- https://data.cyber.org.il/reversing/Inception.exe
- Remove defense using patching

# Classwork

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- https://data.cyber.org.il/reversing/BispooKeygen1.exe

# Small Challenge ☺

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- https://data.cyber.org.il/reversing/GuessPass.exe